

Cross-PDK Validation of a Deterministic Simulator-in-the-Loop Protocol for Standard-Cell Corner Calibration

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Abstract

An offline regression benchmark does not establish that a surrogate reduces new standard-cell SPICE work. We therefore formulate unseen-corner calibration as a deterministic simulator-in-the-loop allocation problem. A controlled GF180MCU study first compares five acquisition policies on 15 feature-only candidate pools. Cell-balanced space filling reaches a median delay R^2 of 0.8729 with 48 of 96 queries, leaving a 0.0120 gap to the measured reference while reducing calls by 50%. The protocol is then tested on released GF180MCU standard-cell CDLs spanning eight logic families, two drive strengths, and three corners. A validation-blind stopping rule is locked on nine GF180MCU development pools and evaluated without retuning on six GF180MCU confirmation pools and 15 external SKY130 pools. It stops at median budgets of 88 and 80, respectively. Five of six GF180MCU runs and all 15 SKY130 runs remain within 0.02 R^2 of their measured 96-call references; the maximum SKY130 gap is 0.0146. Fixed 48- and 64-call budgets meet the same criterion in only 1/15 and 6/15 SKY130 pools. The complete evidence package contains 10,093 successful SPICE calls. A separate 432-point GF180MCU Liberty comparison preserves delay ordering (Spearman $\rho = 0.9900$) while exposing an 18.6% median absolute percentage error. The contribution is a reproducible cross-PDK protocol validation, bounded to PDK-specific retraining and pre-layout exploration rather than zero-shot model transfer, complete Liberty characterization, or

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sign-off.

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1. Introduction

Transistor-level simulation remains the reference for evaluating standard-cell delay and power because these quantities depend jointly on transistor sizing, topology, process corner, voltage, temperature, load capacitance, and input transition time. Even before complete library characterization, a designer may need to explore many combinations to decide which regions deserve a denser sweep. The practical question is therefore not only whether a regression model fits previously generated rows, but whether it can allocate a limited budget of new SPICE simulations without hiding extrapolation failures.

Learning-assisted circuit design has been studied in analog, mixed-signal, and standard-cell settings [1–5]. Bayesian and multi-fidelity methods have further shown that a simulator budget can be treated as part of the optimization problem rather than as a fixed data-collection cost [6–8]. However, three gaps remain important for a small open-PDK study. First, a random split of a static table mainly measures interpolation. Second, pool-based replay does not prove that selected points were actually simulated during acquisition. Third, PDK control parameters can silently change the statistical meaning of the generated data.

The third issue is concrete in the GF180MCU flow used here. The included design configuration enables global and mismatch statistical switches by default. If these switches are not overridden, two nominally identical netlists can produce different values while the resulting table is described as a deterministic PVT study. That behavior may be appropriate for variation analysis, but it confounds a corner-calibration experiment unless random draws and seeds are modeled explicitly. We therefore force both switches off and retain the configuration, model path, netlist, and simulator log for every row.

This paper reframes the study around a genuine simulator-in-the-loop task: a process corner is absent from the base training set; acquisition sees only the features of 96 new candidates; selected points are submitted to

ngspice; and an independent validation set is used only after each update. The same candidate pool is eventually completed to provide a measured reference. The protocol is first transferred from author-defined templates to released GF180MCU standard-cell CDLs and then replicated on released SKY130 cells without retuning the stopping rule. This design supports four contributions:

1. a deterministic GF180MCU/ngspice generation protocol that makes the statistical-mode setting explicit and auditable;
2. a validation-blind unseen-corner support loop with fresh SPICE calls, retained netlists and logs, five acquisition baselines, five seeds, and three held-out corners;
3. a stopping rule developed on released GF180MCU cells, locked before six same-PDK confirmation pools and 15 external SKY130 pools, with every SKY130 pool retained as a measured 96-call reference; and
4. two boundary-setting results: fixed budgets do not transfer reliably across stages or PDKs, and fresh GF180MCU SPICE delay preserves official Liberty ordering despite a measurable absolute bias.

The method is deliberately bounded. It targets early transistor-level exploration of controlled and released combinational-cell arcs. Cross-PDK validation means replication of a fixed allocation and stopping protocol with PDK-specific training data; it does not mean direct transfer of one fitted regression model between technologies. The method does not replace sign-off simulation, layout-parasitic extraction, all timing arcs, leakage-state characterization, or complete Liberty generation.

2. Related work and study position

Machine-learning surrogates are widely used for expensive circuit-response prediction and design-space search [1–3]. In standard-cell characterization, prior work has addressed library modeling, design-technology co-optimization, reliability, knowledge transfer, and graph representations [4, 5, 9, 10]. Those studies motivate learning-assisted characterization. Our complementary focus is the experimental protocol needed to demonstrate that a surrogate actually reduces fresh SPICE work and that a budget selected on controlled circuits remains credible on released library cells.

Sequential design and active learning choose the next expensive label using the current model or the geometry of the unlabeled pool [11]. In

circuit optimization, Bayesian and multi-fidelity methods commonly combine prediction uncertainty, expected improvement, or fidelity-aware cost [6–8]. Our purpose is narrower. We do not claim a new acquisition theory. We test whether a simple policy can be executed as an auditable SPICE loop and how much measured work it saves relative to completing the same candidate pool.

This distinction matters for journal positioning. Model-zoo results are retained only as a same-flow sanity check. The main unit of evidence is a SPICE query, not a row revealed retrospectively from a labeled table. The main comparison is against a measured exhaustive candidate set, not only against another regressor on a random split.

3. Deterministic SPICE protocol

3.1. PDK, simulator, and statistical-mode control

All publication-eligible simulations use ngspice 46 [12]. The GF180MCU stages use the primitive-device model file `sm141064.ngspice` [13–15]. The local PDK checkout corresponds to commit `de3240d`; the primitive-model and 7-track standard-cell submodules correspond to commits `9f992d5` and `43beb45`, respectively [16]. The full hashes are given in the repository references. The controlled templates use `nmos_3p3` and `pmos_3p3`. The released-cell experiment instantiates official cell CDLs and preserves their transistor connectivity, widths, and lengths while mapping their 5-V device symbols to compatible open ngspice model wrappers.

The cross-PDK stage uses the FOSSi Foundation SKY130 primitive-device and high-density standard-cell repositories at commits `403964d` and `9cb2d7c`, respectively [17, 18]. Generated standard-cell SPICE subcircuits instantiate `sky130_fd_pr__nfet_01v8` and `sky130_fd_pr__pfet_01v8_hvt`. Each netlist includes only the corresponding official `tt`, `ff`, or `ss` PM3 models and their mismatch-parameter definitions. The mismatch and process-randomization switches `mc_mm_switch` and `mc_pr_switch` are set to zero, so this stage has the same deterministic estimand as the GF180MCU study.

The GF180MCU design include enables both global and mismatch statistical switches by default. Each generated publication netlist therefore inserts

```
.param sw_stat_global=0 sw_stat_mismatch=0
```

after the design include. This override defines the reported experiment as deterministic corner characterization rather than Monte Carlo variation

analysis. The dataset records `statistical_mode=off`, both switch values, the sampling seed, simulator version, model path, netlist path, log path, fidelity, and completion status.

3.2. Cells, variables, and measured targets

The templates cover controlled arcs for INV, NAND2, NOR2, and XOR2. NAND2 and NOR2 use tied-input transitions. XOR2 holds one input high while the other toggles. The experiment therefore studies controlled topology cases rather than every legal state and timing arc of a production cell library.

The numerical inputs are NMOS width W_n , channel length L , PMOS/NMOS width ratio, supply voltage, temperature, load capacitance, and input slew. Cell type, input arc, and process corner are categorical inputs. The channel length is fixed at $0.28\text{ }\mu\text{m}$. The targets are average propagation delay,

$$d = \frac{t_{\text{PHL}} + t_{\text{PLH}}}{2}, \quad (1)$$

and average switching supply power over the defined transient measurement window. Numerical features are standardized and categorical features are one-hot encoded with unknown-category-safe handling. The controlled experiment assigns each simulated point one of four non-overlapping data roles. Table 1 lists their sizes, label-access rules, sampling functions, and retained provenance.

Table 1: Controlled-stage deterministic datasets and simulator-in-the-loop protocol.

Component	Size	Sampling role	Label access	Provenance
Primary SPICE set	320 rows; 80/cell	Base training and development	Available before online study	Netlist, log, model, switches
Validation SPICE set	480 rows; 120/cell	Independent budget evaluation	Never available to acquisition	Separate sampling seed/directories
Online candidate pool	96 points per seed-corner	Same-corner support search	Features only until selected	Fresh netlist and log per query
Measured reference	96 points per seed-corner	Exhaustive budget reference	Completed after online runs	One retained row per candidate
Seeds and corners	Five pool seeds; held-out ff, ss, and typical corners; four-query batches; maximum online budget 48			
Variable ranges	W_n : 0.517–4.990 μm ; ratio: 1.503–2.999; V_{DD} : 1.620–1.980 V; temperature: -20 – $100\text{ }^\circ\text{C}$; load: 1.066–99.832 fF; slew: 10.672–499.454 ps			

3.3. Independent datasets

The primary dataset contains 320 successful rows generated with Latin hypercube seed 20260530. Each cell contributes 80 rows. The corner counts are 104 ff, 108 ss, and 108 typical rows. The independent validation dataset contains 480 successful rows generated with seed 20260602, with 120 rows per cell and 160 rows per corner. No legacy or externally sourced table is included in the reported training, support, or evaluation sets.

The independent validation set is not cross-PDK or cross-simulator validation. Both datasets use the same PDK, simulator, device models, and

circuit templates. Independence here means a different design-point sample, sampling seed, netlist directory, and log directory within the same controlled flow.

4. Released-library and cross-PDK external validation

4.1. Official CDLs and independent datasets

To test whether conclusions from author-defined templates survive a more realistic topology distribution, we construct a second experiment from the released GF180MCU 7-track standard-cell CDLs. The set contains INV, NAND2, NOR2, XOR2, AOI21, OAI21, MUX2, and NAND3 families at drive strengths 1 and 4, giving 16 cell variants. One sensitized input arc is used per family. For the ngspice flow, official CDL instances of `nfet_05v0` and `pfet_05v0` are mapped to the open `nmos_6p0` and `pmos_6p0` subcircuit wrappers while preserving transistor connectivity, width, and length. This is a released-topology external test within the same PDK, not a claim of bit-identical reproduction of the library provider’s characterization deck.

Independent primary and validation datasets contain 576 successful rows each, generated with seeds 20260717 and 20260731. Each dataset contains 36 points per variant and 192 points per process corner. Besides voltage, temperature, input slew, load, and corner, the model receives family, variant, drive strength, sensitized arc, input count, transistor count, and aggregate NMOS/PMOS width. The joint sampled range is 1.62–1.98 V, -40 – 125 °C, 0.02–12 ns input slew, and 0.001–0.206 pF load.

The online study uses five additional pool seeds and three held-out corners. Each pool contains six unseen points for every variant, or 96 candidates. A variant-balanced farthest-first order is generated from feature geometry, and support is added in 16-query batches so that every variant receives one point per batch. Completing all 15 pools requires 1440 fresh, successful SPICE calls and supplies the measured references used below.

4.2. Validation-blind stopping rule

The released-library stage adds an operational stopping rule that never observes validation labels. After each batch, it evaluates three quantities available during deployment: normalized prequential delay error on the just-measured batch, normalized change in the model’s predictions over the remaining pool, and feature-space coverage radius relative to its initial value. Seeds 0–2 (nine corner pools) are used to choose thresholds. A candidate rule

must keep at least eight of nine development runs within 0.02 R^2 of the full reference and every development run within 0.05. The selected rule requires a minimum budget of 64, prequential normalized MAE no greater than 0.25, prediction change no greater than 0.05, and coverage ratio no greater than 0.50 for two consecutive batches. These thresholds are then locked before evaluation on seeds 3–4 (six confirmation pools).

4.3. SKY130 external protocol replication

The locked rule is next applied to a second open PDK without threshold selection or retuning. The SKY130 high-density set contains the same eight logical families and drive strengths 1 and 4. SKY130 names AOI21 and OAI21 as `a21oi` and `o21ai`; the analysis retains the common logical-family names while recording every original source-cell name and path. One sensitized arc is selected per family using the same logical conditions as in the GF180MCU released-cell study.

Independent SKY130 primary and validation datasets contain 576 successful rows each, generated with seeds 20260718 and 20260819. Each dataset contributes 36 points per variant and 192 points per corner. The sampled voltage, temperature, slew, and load ranges are identical to the released GF180MCU study. Five new pool seeds and three held-out corners provide 15 additional 96-point pools. Acquisition uses the same variant-balanced farthest-first order and 16-query batches. For each corner, the regressor is trained on SKY130 data from the other two corners and updated with newly measured same-corner support. Thus, this experiment tests transfer of the allocation and stopping protocol, not transfer of fitted GF180MCU model parameters.

All 1440 candidate netlists are simulated to obtain measured 96-call references. The stopping decision is then replayed using the unchanged GF180MCU thresholds. The two independent SKY130 datasets and the online pools add 2592 successful calls, bringing the complete two-PDK evidence package to 10,093 successful SPICE calls. Netlists, logs, source-cell paths, PDK commits, model corners, and switch values are retained for audit.

4.4. Partial Liberty compatibility check

As a separate engineering check, released Liberty delay surfaces are compared with fresh simulations of the corresponding released CDLs. For each of 16 variants and three matched PVT files, the low, middle, and high indices

of both the ten-point slew axis and ten-point load axis form nine comparison points, giving 432 simulations. We report rank and absolute agreement separately. This experiment is intentionally partial: it does not reconstruct all arcs, constraints, leakage states, internal power groups, waveform settings, or extracted parasitics of a complete characterization flow.

5. Simulator-in-the-loop corner calibration

5.1. Held-out-corner task

For a held-out corner c , the base set contains only primary rows from the other two corners. A new candidate pool contains 24 Latin-hypercube points per cell, or 96 points in total, all assigned to c . Acquisition initially receives only their input features. When a point is selected, the program writes its transistor-level netlist, launches ngspice, parses delay and power, appends the measured row to the support set, and retrains the surrogate. Validation rows from corner c are then used to record the budget trajectory, but their labels are not passed to acquisition or stopping logic. This information boundary is central to the simulation-saving claim. Fig. 1 traces the label flow and the subsequent locked-rule confirmation and cross-PDK replication sequence.

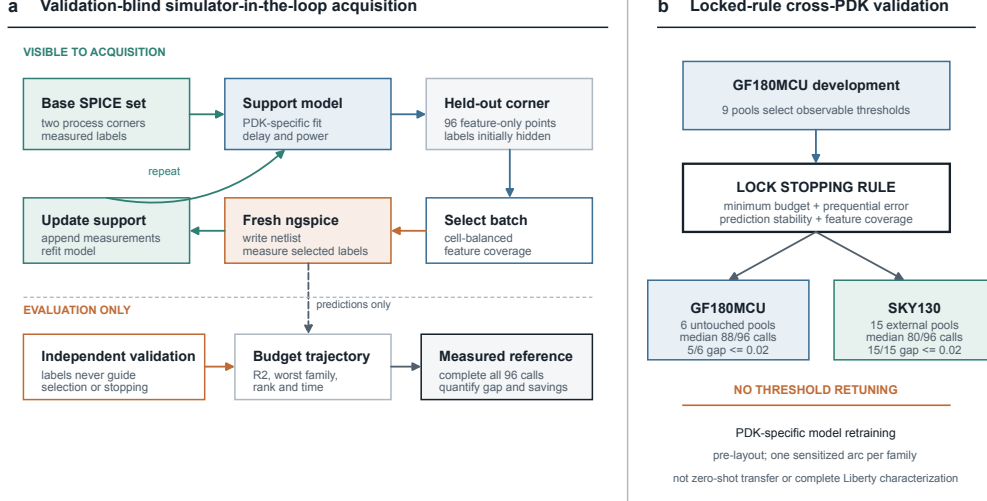


Figure 1: Validation-blind simulator-in-the-loop acquisition and locked-rule cross-PDK validation. **(a)** For a held-out corner, acquisition receives a feature-only 96-point pool. Every selected support label is obtained from a fresh ngspice run, whereas independent validation labels are used only to evaluate the budget trajectory; completing the pool provides the measured exhaustive reference. **(b)** Observable stopping thresholds are selected on nine GF180MCU development pools, locked before six untouched same-PDK confirmation pools and 15 SKY130 external pools are examined, and applied without threshold retuning. Models are retrained within each PDK; the experiment is pre-layout and does not represent zero-shot parameter transfer or complete Liberty characterization.

5.2. Surrogate and acquisition strategies

The online support model is an Extra Trees regressor with 240 trees, `min_samples_leaf=2`, and all transformed features available at each split. Separate models predict delay and power. This model is fixed across all strategies. Its purpose is to provide a stable nonlinear support model, not to establish a new model-ranking result.

For geometric selection, standardized numerical features are concatenated with one-hot cell indicators. Let $g(x)$ denote this vector and S the queried set. The diversity score is

$$q_{\text{div}}(x) = \begin{cases} \min_{s \in S} \|g(x) - g(s)\|_2, & S \neq \emptyset, \\ \|g(x) - \bar{g}\|_2, & S = \emptyset, \end{cases} \quad (2)$$

where $\bar{g}$ is the candidate-pool centroid. Each four-query cell-balanced space-filling batch selects one available point from each cell with the largest current diversity score. A primary-data-only development study fixed the policy to pure diversity before the independent online evaluation.

Five policies use identical pools: random selection, cell-balanced random selection, uncertainty-only selection from tree-ensemble disagreement, ordinary greedy space filling, and cell-balanced space filling. The last policy is the operational method studied here. It guarantees equal support counts across the four cells at every completed batch. We do not assume that it must outperform ordinary space filling; the comparison tests whether cell balance changes reliability or cost.

5.3. Evaluation and statistics

After budgets $b \in \{0, 4, \dots, 48\}$, the model is evaluated on all 160 validation rows from the held-out corner. We report delay and power R^2 , MAE, the minimum delay R^2 across the four cells, Spearman correlation, Kendall’s tau, top-20% precision and recall, and NDCG. The main metric is independent-validation delay R^2 because delay shows the strongest zero-shot corner failure.

The complete design has five seeds, three held-out corners, and five strategies, yielding 75 online runs and 3600 actual ngspice queries. Seventy-seven additional calls complete candidates not selected by any strategy. The online and completion phases therefore contain 3677 new simulator calls and 2311.5 s of measured serial-equivalent wall time.

At budgets 24 and 48, the cell-balanced space-filling policy is compared with each baseline over 15 matched seed-corner pairs using two-sided Wilcoxon signed-rank tests. Holm correction is applied within each budget-metric family. The paired design prevents differences among candidate-pool seeds or corners from being mistaken for acquisition effects.

6. Results

6.1. Configuration control restores numerical repeatability

The first result is methodological rather than predictive. Before the statistical-mode override, repeated nominal candidate simulations differed because global and mismatch variation were enabled by the included PDK configuration. With both switches forced to zero, independently launched repeats are identical. Across the online study, 1118 candidate groups were

queried by at least two strategies, accounting for 3355 query records. The maximum within-group span is exactly 0 for t_{PHL} , t_{PLH} , average delay, and average power at the stored precision. This audit prevents stochastic PDK variation from being misreported as deterministic PVT response or acquisition noise.

As a secondary same-flow sanity check, models trained on all 320 primary rows generalize to the independently sampled 480-row validation set. CatBoost reaches delay $R^2 = 0.9208$ and MAE 0.0321 ns, while Gaussian-process regression reaches power $R^2 = 0.9994$ and MAE 0.0449 μW . These values show that the deterministic tables are learnable, but they are not the main evidence for simulation reduction.

6.2. Same-corner queries repair delay extrapolation

Without same-corner support, the cell-balanced space-filling trajectory has pooled median delay $R^2 = 0.4825$ over the 15 seed-corner pairs. The corner dependence is substantial: median zero-support R^2 is 0.3055 for ff, 0.4825 for ss, and 0.7877 for typical. This confirms that a strong random-split or same-flow validation score does not justify zero-shot use at an absent corner.

Fresh same-corner SPICE labels improve the difficult corners. At 24 queries, median delay R^2 reaches 0.7763 for ff, 0.8035 for ss, and 0.8896 for typical. At 48 queries, the values are 0.8797, 0.8488, and 0.8887. The pooled worst-cell delay R^2 rises from 0.3112 without support to 0.6785 at 24 queries and 0.7885 at 48 queries. Power changes little because its zero-support transfer is already strong: pooled median power R^2 is 0.9455 at zero support and 0.9465 at 48 queries. Table 2 resolves these trends by held-out corner and compares each support budget with its measured 96-query reference and simulation time.

6.3. Space coverage is safer than uncertainty-only acquisition

To separate acquisition-policy effects from the benefit of support volume alone, Fig. 2(a-c) compares matched budget trajectories. At 24 queries, cell-balanced space filling has pooled median delay $R^2 = 0.8119$, compared with 0.7625 for random, 0.7812 for cell-balanced random, 0.8114 for ordinary space filling, and 0.7232 for uncertainty-only selection. The difference from uncertainty-only selection remains significant after Holm correction ($p = 0.0171$). The difference from random is positive but does not cross the corrected 0.05 threshold ($p = 0.0646$).

Table 2: Cell-balanced space-filling results by held-out corner. Values are medians over five candidate-pool seeds.

Corner	Support	Delay R^2	Worst-cell delay R^2	Top-20% recall	Measured time (s)
ff	0	0.3055	-0.1887	0.8750	0.00
	24	0.7763	0.6000	0.8750	14.40
	48	0.8797	0.7883	0.8750	28.88
	96 reference	0.9054	0.8539	0.8750	59.62
ss	0	0.4825	0.3112	0.9688	0.00
	24	0.8035	0.7060	0.9688	14.38
	48	0.8488	0.7925	0.9688	28.86
	96 reference	0.8541	0.7853	0.9688	59.20
typical	0	0.7877	0.4872	0.8125	0.00
	24	0.8896	0.7755	0.8750	14.40
	48	0.8887	0.7983	0.9062	28.87
	96 reference	0.8824	0.7918	0.9375	58.73

At 48 queries, the respective medians are 0.8729, 0.8564, 0.8685, 0.8674, and 0.8111. Cell-balanced space filling exceeds random ($p = 0.0374$) and uncertainty-only selection ($p = 0.0334$) after correction. It is statistically indistinguishable from cell-balanced random and ordinary space filling ($p = 1.0$ for both corrected comparisons). The worst-cell advantage over random is not significant after correction ($p = 0.1414$). These tests support a restrained conclusion: covering the new corner is safer than relying only on ensemble disagreement, while the present data do not demonstrate superiority over all space-filling alternatives.

Cell balance is nevertheless operationally explicit. Its support-count range is zero at every complete four-query batch. At 48 queries, the median range is four points for random, ordinary space filling, and uncertainty-only selection. This property prevents one cell from consuming most of a small support budget even when aggregate accuracy appears acceptable. Fig. 2(d) then places the selected 24- and 48-query budgets against the distribution of measured 96-query references.

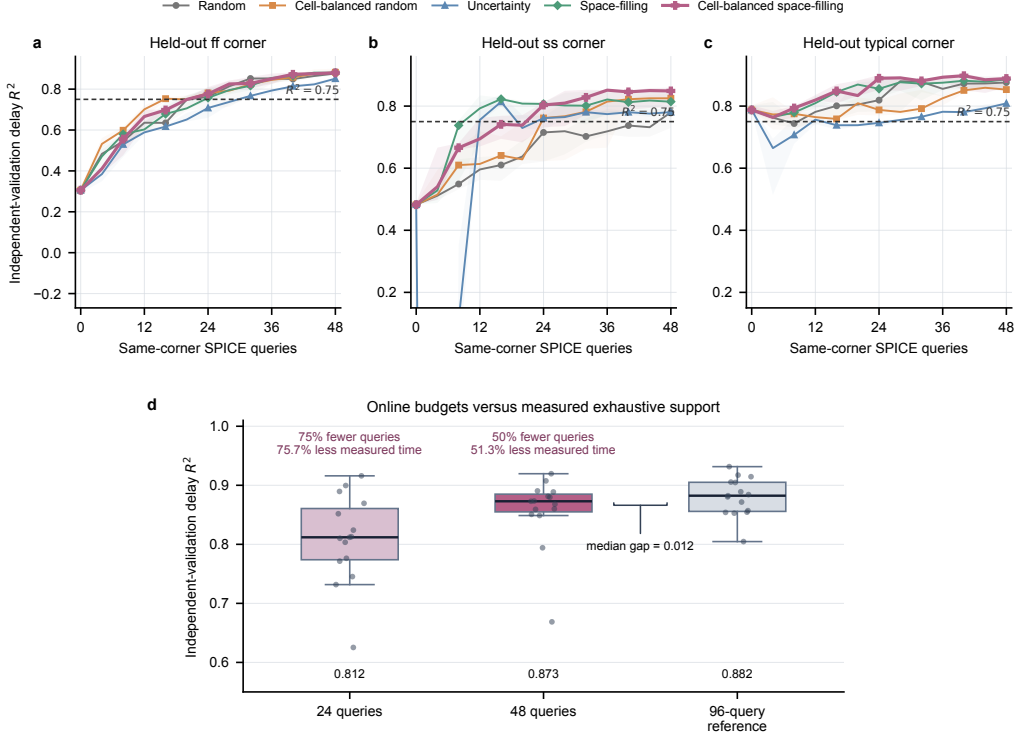


Figure 2: Validation-blind simulator-in-the-loop results. (a–c) Median independent-validation delay R^2 across five seeds for each held-out process corner; translucent bands show the interquartile range. The horizontal line is the prespecified $R^2 = 0.75$ study threshold. (d) Distribution over 15 matched seed-corner runs for cell-balanced space filling at 24 and 48 queries and for the measured 96-query reference. Each point is one seed-corner pair. The 48-query budget reduces SPICE calls by 50% and measured time by 51.3%, with a median R^2 gap of 0.012 to the full reference.

6.4. Half of the measured candidate pool approaches the exhaustive reference

Completing every candidate produces 15 measured 96-query references, one for each seed-corner pool. Their pooled median delay R^2 is 0.8824, median worst-cell delay R^2 is 0.8150, and median top-20% recall is 0.9375. With 24 queries, cell-balanced space filling uses 75% fewer calls and 75.7% less measured time; median delay R^2 is 0.8119 and the median gap to the reference is 0.0798. With 48 queries, call reduction is 50%, time reduction is 51.3%, median delay R^2 is 0.8729, and the median gap is 0.0120.

The 48-query budget also preserves the reference median top-20% recall of 0.9375. Its median worst-cell delay R^2 is 0.7885, compared with 0.8150 for

Table 3: Online budgets compared with the measured 96-query reference over 15 seed-corner runs.

Support	Query reduction	Time reduction	Delay R^2	Gap to reference	Worst-cell R^2	Top-20% recall
24 queries	75.0%	75.7%	0.8119	0.0798	0.6785	0.8750
48 queries	50.0%	51.3%	0.8729	0.0120	0.7885	0.9375
96-query reference	0	0	0.8824	0	0.8150	0.9375

the reference. These results quantify the engineering trade-off directly: half of the candidate simulations retain most of the measured support benefit, whereas one quarter of the pool is useful but leaves a larger accuracy gap. Table 3 consolidates the resulting call, time, accuracy, and ranking-recall trade-offs.

6.5. Validation-blind prequential error is informative but not a stopping guarantee

For each four-query batch, predictions are recorded before its SPICE labels are added. This prequential error is observable during operation and does not use the independent validation labels. Across 180 batches, prequential delay MAE correlates negatively with external delay R^2 (Spearman $\rho = -0.3869$, $p = 8.09 \times 10^{-8}$) and external worst-cell delay R^2 ($\rho = -0.3352$, $p = 4.22 \times 10^{-6}$). The direction is useful, but the magnitude is only moderate. We therefore treat prequential error as a diagnostic for suspicious batches, not as a validated automatic stopping rule.

6.6. The protocol transfers to released standard-cell CDLs

All 1152 independently generated released-CDL rows and all 1440 on-line candidate simulations complete successfully with the statistical switches disabled. The released-library task is materially harder than the controlled four-topology stage. Across 15 seed-corner pools, median delay R^2 is 0.2171 without target-corner support, 0.8258 at 32 queries, 0.8684 at 48, 0.9028 at 80, and 0.9070 after all 96 candidates are measured. Median worst-family R^2 rises from -0.1009 at zero support to 0.3818 at 48 and 0.6932 at 96. Table 4 reports the pooled budget trade-off, while Fig. 3(a) resolves the trajectory by held-out corner.

This transfer test changes the interpretation of the controlled result. A fixed 48-query budget is within 0.02 of the reference in only one of six untouched confirmation pools; its median gap is 0.0427 and maximum gap is 0.1219. The validation-blind rule instead stops at a median of 88 queries.

Table 4: Released-CDL external-validation budgets over 15 seed-corner pools. Values are pooled medians.

Support	Delay R^2	Gap to reference	Worst-family R^2	Measured time (s)
0	0.2171	0.6863	-0.1009	0.00
32	0.8258	0.0986	0.1697	61.09
48	0.8684	0.0451	0.3818	92.58
80	0.9028	0.0106	0.6489	157.25
96 reference	0.9070	0	0.6932	185.60

Five of six confirmation runs are within 0.02, all six are within 0.05, and the maximum gap is 0.0207. The corresponding median delay R^2 is 0.9036 and median worst-family R^2 is 0.7348. Fig. 3(b) shows that the rule is conservative, with a median query reduction of only 8.3%. It nevertheless avoids presenting the successful controlled-stage half-budget as a universal constant.

Fig. 3(c) tests a different question: whether fresh ngspice simulations preserve the ordering and magnitude of released Liberty delays. Spearman $\rho = 0.9900$, Pearson $r = 0.9916$, and $R^2 = 0.9460$ indicate strong agreement in ordering and linear association. However, the median absolute percentage error is 18.58%, and the median SPICE-to-Liberty delay ratio is 0.814. The ordering agreement supports early ranking, whereas the absolute bias confirms that the present deck does not replace the provider’s complete characterization conditions.

6.7. The GF180MCU stopping rule transfers to SKY130 without retuning

The decisive transfer question is whether the GF180MCU-locked rule controls error better than fixed budgets on SKY130. Table 5 summarizes this matched comparison across both PDKs. The prelocked rule satisfied the primary 0.02-gap criterion in all 15 SKY130 pools. Its median stop was 80 of 96 calls, corresponding to a 16.7% median reduction, and its largest delay R^2 gap to a measured reference was 0.0146. The median stopped-model delay R^2 was 0.9086, and the median worst-family R^2 was 0.7235. Seven pools, including all five ss pools, continued to the complete 96-call reference; the rule therefore saved work only when its observable stability and coverage conditions were met.

Fixed budgets did not provide the same protection. A 64-call budget met the 0.02 criterion in 6 of 15 SKY130 pools and produced a maximum gap of 0.0945. A 48-call budget met it in only 1 of 15 pools and produced a

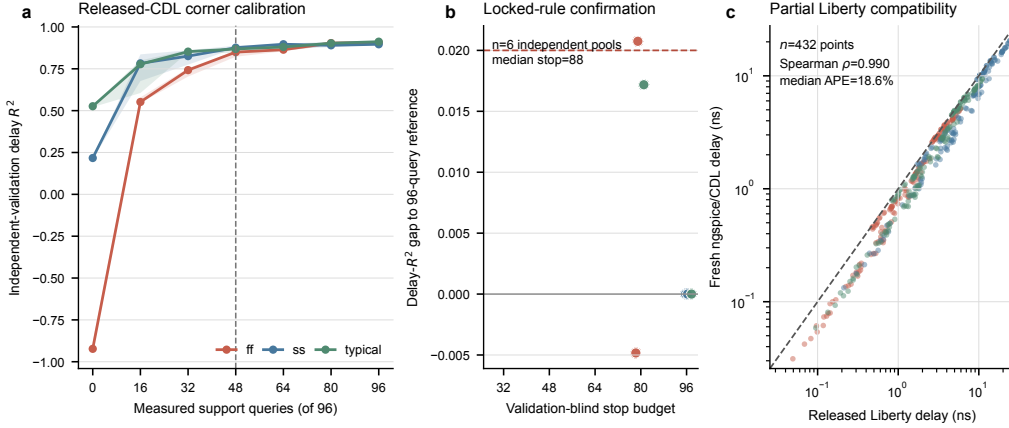


Figure 3: External validation on released GF180MCU standard-cell CDLs. (a) Median independent-validation delay R^2 versus measured support for the three held-out corners; bands show the interquartile range over five pool seeds. (b) Untouched confirmation pools compare the validation-blind stopping budget with the resulting gap to each measured 96-query reference. Dashed lines mark gaps of 0.02 and 0.05. (c) Fresh ngspice delay versus official Liberty delay at 432 matched cell-PVT-slew-load points. The log-log identity line exposes the absolute offset while rank correlation quantifies ordering agreement.

maximum gap of 0.1516. The same ordering is present in the six untouched GF180MCU confirmation pools: the locked rule succeeds in five, whereas each fixed budget succeeds in one. To determine whether the aggregate rates hide corner-specific failures, Fig. 4 shows every locked-rule outcome and the matched fixed-budget comparisons. The result supports transfer of the stopping protocol across these two PDKs, but it does not imply that a regressor fitted on GF180MCU directly predicts SKY130 cells.

Table 5: Stopping-rule replication across released GF180MCU and SKY130 standard cells. Success means a delay R^2 gap no greater than 0.02 relative to the measured 96-call reference.

PDK	Method	Pools	Median calls	Median reduction	Successful pools	Maximum gap
GF180MCU	Locked rule	6	88	8.3%	5/6	0.0207
GF180MCU	Fixed 64	6	64	33.3%	1/6	0.0537
GF180MCU	Fixed 48	6	48	50.0%	1/6	0.1219
SKY130	GF180MCU-locked rule	15	80	16.7%	15/15	0.0146
SKY130	Fixed 64	15	64	33.3%	6/15	0.0945
SKY130	Fixed 48	15	48	50.0%	1/15	0.1516

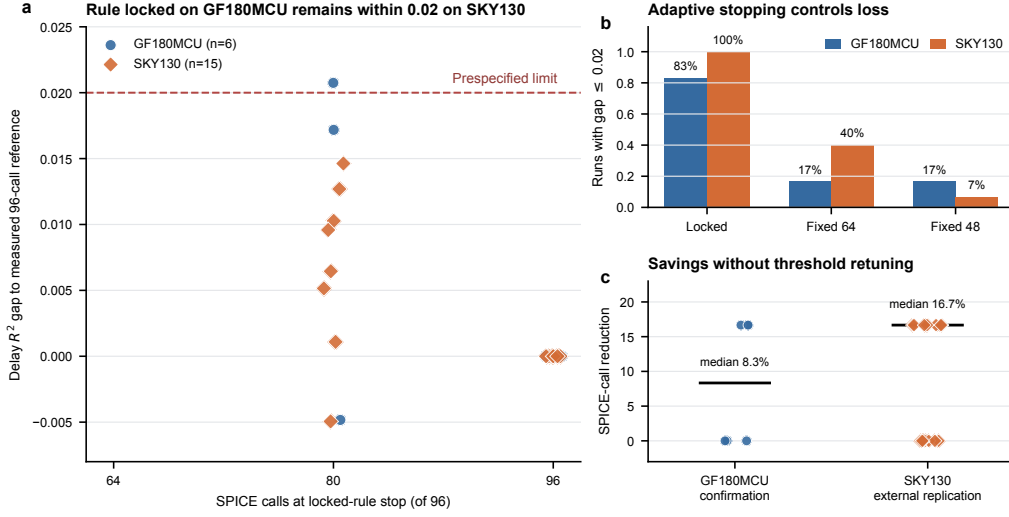


Figure 4: Cross-PDK replication of the validation-blind stopping rule. (a) Individual outcomes for the six untouched GF180MCU confirmation pools and 15 external SKY130 pools. The rule was selected using separate GF180MCU development pools; the dashed line is the prespecified 0.02 delay- R^2 gap limit. Negative gaps indicate that an early model slightly outperformed the corresponding 96-call model on the independent validation set. (b) Fraction of pools within the 0.02 limit for the locked rule and fixed 64- and 48-call budgets. (c) Call reduction for each locked-rule outcome; horizontal bars show medians. SKY130 thresholds were not retuned, and all candidate pools were completed to obtain measured references.

7. Discussion

7.1. What changes relative to an ML benchmark

The central result is no longer that one regressor achieves a larger random-split score. Every online support label is produced by a retained simulator call, validation labels are isolated from acquisition, and budget savings are measured against the same pool completed by SPICE. The controlled stage establishes repeatability and matched-policy behavior. The released GF180MCU stage tests larger topology and drive-strength diversity, while SKY130 asks whether the locked operational rule survives a change of device models, cell implementations, and process node. This evidence chain addresses simulation allocation under an unseen-corner stress test rather than static tabular prediction.

The configuration audit is equally important. Open PDKs improve reproducibility only when their operating controls are recorded. Global and

mismatch variation should not be disabled universally; they are essential for variation analysis. They are disabled here because the estimand is deterministic corner response. A future statistical-characterization study should instead expose random seeds, repeat counts, and distributional targets as first-class variables.

7.2. Why uncertainty-only selection underperforms

Tree disagreement is conditional on the current training distribution. At an unseen corner, early uncertainty estimates can be poorly calibrated and can repeatedly favor a narrow region. The sharp early degradation visible in some uncertainty-only trajectories is consistent with this failure mode. Geometric coverage does not require the surrogate to be calibrated before the first target-corner labels arrive. Cell balancing adds a simple topology-level safeguard.

This finding should not be generalized to all uncertainty acquisition. Gaussian-process posteriors, calibrated ensembles, density weighting, or multi-fidelity acquisition may behave differently. The result only shows that a common tree-ensemble disagreement heuristic is not automatically reliable in this small corner-transfer setting.

7.3. Practical interpretation

The 96-point candidate pool is an early-exploration reference, not a complete characterization grid. The measured time per controlled 96-query reference is about 59 s, whereas the released-CDL pools have a median of 185.6 s because they contain larger official topologies and broader slew/load ranges. Production characterization would involve more arcs, loads, slews, states, corners, and extracted views, so absolute cost would be much larger. The defensible extrapolation is qualitative: a validation-blind support loop can decide where to spend a bounded number of new simulations, while final selected conditions remain subject to the intended sign-off flow.

The three stages also show why a fixed budget should not be elevated into a general rule. Forty-eight queries are effective for the controlled templates but unreliable for both released libraries. The locked stopping rule is intentionally conservative: its median saving is 8.3% in GF180MCU confirmation and 16.7% in SKY130, and it performs all 96 calls whenever its stability conditions are not met. This behavior gives up aggressive savings to improve gap control without consulting validation labels. A deployment with a different

loss tolerance should develop new thresholds before its confirmation data are examined rather than reinterpret the present values as universal constants.

7.4. Limitations

The study uses two open PDKs, one simulator version, and pre-layout netlists. Both released-library stages cover eight combinational families and two drive strengths, including AOI21, OAI21, and MUX2, but evaluate only one sensitized arc per family. They do not cover sequential cells, every input-state arc, leakage and internal-power tables, timing constraints, or extracted parasitics. Each PDK uses its own training and validation data, so the experiment establishes cross-PDK protocol replication rather than zero-shot model transfer. Generalization to a third PDK, a different simulator, or a layout-extracted flow remains untested.

The Liberty comparison is a 432-point GF180MCU surface compatibility check, not full characterization, and no matched SKY130 Liberty comparison is claimed. Its 18.58% median absolute percentage error indicates that waveform, deck, and parasitic differences still matter despite excellent rank agreement. The acquisition policy is deliberately simple and is not claimed as algorithmically novel. One of six GF180MCU confirmation pools exceeds the 0.02 gap by 0.0007, although all remain within 0.05; the 15 SKY130 pools satisfy the primary threshold but still constitute a finite cohort generated with one sampling design. Larger locked cohorts and post-layout tests are needed before industrial characterization claims are warranted.

8. Conclusion

This paper presents a deterministic, provenance-controlled simulator-in-the-loop protocol for unseen-corner support and validates its stopping behavior across GF180MCU and SKY130. The controlled stage shows that geometry-aware acquisition can approach a measured reference with half of a 96-call pool. The released-cell stages provide the decisive evidence: a rule locked on GF180MCU stops at median budgets of 88 in six GF180MCU confirmation pools and 80 in 15 external SKY130 pools. Five of six and 15 of 15 runs, respectively, remain within 0.02 delay R^2 of their measured references, whereas fixed budgets fail more often. The complete evidence package contains 10,093 successful SPICE calls. A 432-point GF180MCU check additionally preserves official Liberty delay ordering ($\rho = 0.9900$) while exposing an 18.58% median absolute error. The contribution is therefore an

auditable simulation-allocation and stopping protocol with cross-PDK replication, bounded to PDK-specific retraining and pre-layout exploration rather than zero-shot transfer, sign-off, or complete Liberty generation.

CRediT authorship contribution statement

Jiaqing Chen: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization. Feng Shao: Supervision, Conceptualization, Methodology, Writing – review and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data and code availability

Data and code supporting this study are archived in the versioned Zenodo record <https://doi.org/10.5281/zenodo.21415175>. The corresponding live repository is available at <https://github.com/jqchen0715/gf180mcu-spice-surrogate-mej>. Release v0.5.0 contains the GF180MCU and SKY130 released-cell primary and validation data, 2880 online query records, cross-PDK stopping decisions, the 432-point GF180MCU Liberty comparison, retained per-query netlists and ngspice logs, exact PDK commits, analysis scripts, numeric audits, and figure source data.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used AI-assisted tools for manuscript organization, language editing, code assistance, and consistency checking. The authors reviewed and edited all outputs and take full responsibility for the content of the article.

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